

<u>Discrete</u>	<u>scenario</u>	<u>support</u>
Bernoulli (p)	1 success / failure	0, 1
Binomial n, p	# of successes (in n trials)	0, 1, ..., n
Geometric p	# of trials until 1 st success	1, 2, ...
Neg. Binom r, p	# of trials until r^{th} success	$r, r+1, \dots$
Hypergeometric r, n, N	# of units w/ characteristic (out of r) in sample of size n from finite pop of size N	$y \leq r$ $y \leq n$
Poisson λ	# of occurrences in an interval	0, 1, 2, ...
Uniform	anything that has constant prob. rolling dice	1, ..., m

Special cases : Bernoulli is special case of Binomial w/ $n=1$
 geometric is " of Neg. Binom w/ $r=1$

Continuous

if I have pdf $f(x)$

$$F(x) = \int_{-\infty}^x f(t) dt$$

if I have cdf $F(x)$

$$f(x) = F'(x)$$

$$F(x) = \begin{cases} 0 & \text{when } x < lb \\ \frac{F(x) - F(lb)}{F(ub) - F(lb)} & \text{when } lb < x < ub \\ 1 & \text{when } x > ub \end{cases}$$

don't forget to define support when writing $f(x), F(x)$

Uniform
 θ_1, θ_2

constant



$$\theta_1 < y < \theta_2$$

Normal
 μ, σ^2



test scores
height, bp

$$-\infty < y < \infty$$

Exponential
 $\beta = \frac{1}{\lambda}$

wait time until 1st occur.

$$0 < y < \infty$$

Gamma
 β, α

wait time until α^{th} occur.

$$0 < y < \infty$$

Chi-square
 $r = df$

used in regression, ANOVA,
hypothesis testing
special case of gamma w/ $\alpha = r/2$
 $\beta = 2$

$$0 < y < \infty$$

Beta

proportions

$$0 < y < 1$$

Exponential is a special case of gamma with $\alpha = 1$

Uniform is special case of beta w/ $\alpha = 1, \beta = 1$

$$Z \sim N(0, 1) \rightarrow Z^2 \sim \chi^2_1$$

$$\Gamma(n) = (n-1)!$$

$$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha + \beta)}$$

1 st moment	$E(x)$	$E(x) = \sum x p(x)$	$E(x) = \int x f(x) dx$
2 nd moment	$E(x^2)$	$\sum x^2 p(x)$	$\int x^2 f(x) dx$
3 rd moment	$E(x^3)$		
	$\vdots$		
k th moment	$E(x^k)$	$\sum x^k p(x)$	$\int x^k f(x) dx$

$$M(t) = E(e^{tx}) = \int e^{tx} f(x) dx \quad \text{or} \quad \sum e^{tx} p(x)$$

$$M'(t=0) = E(x)$$

$$M''(t=0) = E(x^2)$$

$$\vdots$$

$$V(x) = M''(t=0) - (M'(t=0))^2$$

$$V(x) = E[(x-\mu)^2] = 2^{\text{nd}} \text{ central moment}$$

how can e^x be written as inf. series

$$\sum_{y=0}^{\infty} \frac{x^y}{y!} = e^x$$

$$\sum_{n=0}^{\infty} \frac{x^n}{n!} = e^x$$

$$(a+b)^n = \sum_{y=0}^n a^y b^{n-y}$$